

MEASURING ORTHOGONALITY BETWEEN PAIRED RESERVOIRS: DUPLICATE TWINS ARE REPARAMETRIZATIONS, INDEPENDENT TWINS ARE NEARLY FREE

A P R E P R I N T

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ABSTRACT

An open-ended design brief called for two reservoirs, one "orthogonal" to the other, leaving open what orthogonal should mean; this study makes it a measured quantity. Two half-width reservoirs (N=512 each) run in parallel on the identical text stream, states concatenated to width 1024; the second reservoir — the twin — is either a *bit-identical duplicate* of the first or an *independent draw*; orthogonality is measured by canonical correlation between the two 512-dimensional blocks. Two findings, each obtained on all three seeds, and one registered methodological observation. (1) A duplicate twin is a pure reparametrization, **exactly**: it ties the lone half-width reservoir on ridge bpc and decode depth to the last recorded digit, three seeds in a row. (2) An independent twin beats the duplicate on **every** instrument — ridge, logistic, and decode depth — and lands within 0.0008–0.0049 bits of a full-width 1024-unit reservoir built as a single unit: near-free doubling. (3) The observation: the naive reading of the orthogonality measure is misleading — the two reservoirs' *single best-aligned* directions correlate above 0.999 whether or not the wiring was copied, because shared input dominates the top of the spectrum; only the *full-spectrum* mean canonical correlation (≈ 0.90 duplicate vs ≈ 0.45 independent at full budget, recorded on two of the three seeds) distinguishes a copy from a genuinely independent reservoir, and predicts which one pays downstream.

Keywords reservoir computing · canonical correlation analysis · redundancy · ensembles · feature width · pre-registration

§1 Introduction

When does a second reservoir add information to a first one? Folk intuition says: when it is "orthogonal" — differently wired, independently random. But two reservoirs driven by the same input are not independent systems; both spend most of their capacity linearly tracking the same highly-recoverable recent-character history. The brief left the interpretation of orthogonality open; the interpretation adopted here is an *operationalized* one — pick a measure, then check whether the measure predicts the downstream payoff of concatenating the two reservoirs' states.

§2 Method

Two half-width reservoirs built from the program's reference sparse-mixing tanh construction (N=512 each, spectral radius $\rho=0.95$, fanin 10, the standard input map, leak 1.0) [1] run in parallel on the identical text8 stream [3], 2M train characters; their states are concatenated to readout width 1024, matched to the standard full-width reservoir of the addressed-memory ESN study so no feature-width confound enters. Three cells per seed: **single512**, one half-width reservoir alone; **twin_redundant**, in which the twin — the paired second reservoir — is built from the first reservoir's own seed: identical wiring, bit-identical states every step, the maximally non-orthogonal case; **twin_orthogonal**, in which the twin is an independent draw. Orthogonality is computed, not assumed: canonical correlation analysis [2] between the two 512-dim state blocks on subsampled validation rows ($\leq 100k$, ridge-regularized whitening), summarized two ways — `top_canon_corr`, the single best-aligned direction, and `mean_all_canon_corr`, the average over all 512 canonical directions. The full-width comparator (ridge 3.1472, logistic 2.6727, depth 10.11) is the archived reference cell, cited from the archive and not rerun. Instruments: ridge and logistic validation bits per character (bpc), and U3 decode depth — a linear decode-depth probe, so designated in the program's internal instrument numbering.

§3 Pre-registered predictions

The pre-registration pilot run held a genuine surprise, disclosed in the registration before the full run: *independent draws do not decorrelate the top of the spectrum*. The predictions were written with that disclosed:

- **O1/O2** (manipulation checks): the full-spectrum mean separates the twins by a wide margin, while the top canonical correlation stays ≥ 0.99 in *both* twin arms — with the named alternative that more data at full budget resolves the saturation away, in which case it is an artifact of the pilot run. (Both hit; the saturation is real at $20\times$ the data.)
- **O3** (reparametrization null): the duplicate twin ties the lone half-width reservoir on ridge bpc ($|\text{gap}| \leq 0.01$) and decode depth ($|\text{gap}| \leq 0.3$). (Hit — exactly, gap 0.0000 on both.)
- **O4** (the headline): the independent twin beats the duplicate on all three of ridge, logistic, and depth. Named alternative: any single reversal = the pilot run's pattern was small-sample noise. (Hit, 3/3.)
- **O5** (near-free doubling): the independent twin lands closer to the full-width comparator than the lone half-width reservoir does, on both bpc instruments. (Hit.)

Quantitative bands Q1–Q4 were registered separately; two hit and two missed *in the beat-expectations direction* — the 100k-character pilot run undersold how far the logistic readout converges at 2M characters, so both logistic bands' actuals came in below their registered floors. Recorded as misses, consistent with this program's observed pattern of quantitative bands missing. The seed-1 rerun registered O3/O4/O5 unbundled (each promoting alone); a third seed was subsequently commissioned and run under the identical registered wording.

§4 Results

Table 1. All three seeds (internal run log #18 / #19 / #25; values seed 0 / 1 / 2). Full-width comparator, cited from the archive and not rerun: ridge 3.1472, logistic 2.6727, depth 10.11. Zero consistency-gate failures, and no activation of the divergence tripwire (the program's fit-invalidating divergence rule).

cell	ridge val bpc	logistic val bpc	U3 depth
single512	3.3142 / 3.3100 / 3.3013	2.8356 / 2.8318 / 2.8266	8.66 / 8.57 / 8.18
twin_redundant	3.3142 / 3.3100 / 3.3013	2.8330 / 2.8289 / 2.8229	8.66 / 8.57 / 8.18
twin_orthogonal	3.1442 / 3.1487 / 3.1423	2.6713 / 2.6735 / 2.6714	10.08 / 10.18 / 10.03

The duplicate is inert, to the digit. On every seed, twin_redundant's ridge bpc and decode depth equal single512's exactly as recorded — 3.3142/8.66, 3.3100/8.57, 3.3013/8.18 — the cleanest null this program has produced. A duplicated readout column adds no information the ridge instrument or the decode-depth probe can use; this is the 2×-redundancy edge case of the program's measured result that widening a readout in this way is a reparametrization, now with an exact copy instead of a permuted view.

The independent twin wins on every instrument, three times. Ridge 3.1442/3.1487/3.1423 vs the duplicate's 3.3142/3.3100/3.3013; logistic 2.6713/2.6735/2.6714 vs 2.8330/2.8289/2.8229; depth 10.08/10.18/10.03 vs 8.66/8.57/8.18. And it sits 0.0030/0.0015/0.0049 ridge-bits and 0.0014/0.0008/0.0013 logistic-bits from the full-width comparator, where the lone half-width reservoir sits 0.15–0.17 bits away on both — at least thirty times closer, on every seed and instrument. Two independently drawn half-width reservoirs recover almost everything a single full-width reservoir obtains.

The top-direction summary saturates. top_canon_corr: 0.9999 (duplicate, every seed) vs 0.9995/0.9994/0.9995 (independent) — indistinguishable in practice. The full-spectrum mean separates them decisively: 0.9060/0.9022 vs 0.4477/0.4441 on the two seeds where the full-budget run recorded it (the third seed's verdict table recorded the top-direction summary only; its pilot run showed the same shape, 0.8967 vs 0.4562). Registered as a methodological observation and carried in the program's intake notes rather than its confirmed-findings record: it is a point about how to *measure* orthogonality between co-driven reservoirs, not an effect size.

Why the duplicate's full-spectrum mean is 0.906 and not 1.0. A reader may reasonably expect any whole-spectrum correlation summary between an exactly identical pair to read 1.0. For this regularized construction it does not, and the shortfall is closed-form rather than empirical. The estimator adds a ridge term $\text{reg} \cdot \mathbf{I}$, here 10^{-3} , to both auto-covariances before whitening, but not to the cross-covariance. For a bit-identical pair the three covariances are the same matrix $\mathbf{C}$, so the whitened matrix is $(\mathbf{C} + \text{reg} \cdot \mathbf{I})^{-1/2} \mathbf{C} (\mathbf{C} + \text{reg} \cdot \mathbf{I})^{-1/2}$, which shares $\mathbf{C}$'s eigenvectors and has eigenvalues $\lambda_i / (\lambda_i + \text{reg})$ — strictly below 1 for every finite λ_i . A follow-up (internal run log #50) confirmed the pair is bit-identical by direct array comparison at seed 0 (maximum absolute difference exactly 0.0) and checked the prediction numerically: the predicted eigenvalue ratios match the estimator's actual singular values to 4.4×10^{-14} , and the predicted mean reproduces the recorded 0.9060 to all four decimals. The raw covariance spectrum runs from 0.00117 to 14.15, with 212 of 512 eigenvalues below 10^{-2} , ten times the ridge term — the regime where the suppression bites — which is consistent with a 512-unit reservoir driven by a 27-symbol alphabet carrying a long tail of low-variance, near-redundant directions. Sweeping the ridge term down confirms the limit: the predicted mean rises to 0.999885 at $\text{reg}=1\text{e-}6$ and to 1.000000, at displayed precision, at $\text{reg}=1\text{e-}9$. The shortfall from 1.0 is entirely and only a function of the regularization, not evidence of any difference between the two blocks. This paper

makes no claim that the statistic *should* read 1.0; it is used throughout as a discriminating quantity (0.906 against 0.448) and that use is unaffected. The check described here is an internal validity check, not the independent audit this finding still awaits.

§5 Discussion

Duplication is exactly worthless; independence is nearly free. If a second reservoir is to add value it must be drawn independently; doing so buys almost the whole doubling: two half-width reservoirs recover a full-width reservoir to within thousandths of a bit, three seeds in a row. Copying the wiring buys exactly nothing, to the recorded decimal, three seeds in a row. The result composes with the partitioned sub-reservoir ensemble finding from the same day's program — there, 64 blocks of a single draw at matched width never cost and usually helped; here, two fully independent draws (own mixing matrix *and* own input map) nearly match a monolithic reservoir.

"Orthogonal" is a measurement, not a construction. Two reservoirs with independent wiring *sound* orthogonal, but their best-aligned directions correlate above 0.999, because the shared input dominates what both reservoirs most strongly represent. Independence only shows up — and only pays — across the rest of the spectrum. Any pipeline that certifies diversity by construction ("different seeds, therefore decorrelated") is measuring the wrong end of the canonical spectrum.

§6 References

1. H. Jaeger, "The 'echo state' approach to analysing and training recurrent neural networks," GMD Report 148, German National Research Center for Information Technology, 2001.
2. H. Hotelling, "Relations between two sets of variates," *Biometrika* 28, 1936.
3. M. Mahoney, "Large text compression benchmark" (text8), mattmahoney.net/dc/textdata.

§7 Provenance

- **c660a423** — the initiating design brief; 3 cells, seed 0 (internal run log #18). Registered before running; the pilot run's saturation surprise disclosed in the registration. Headline hit 3/3 ⇒ provisional, replication auto-queued before any confirmed-findings entry.
- **142d0104** — seed 1, fresh independent draws of both reservoirs (internal run log #19). All three unbundled claims hit; promoted.
- **8879faae** — seed 2, commissioned as a direct third-seed confirmation; identical registered wording (internal run log #25). All three claims hit again; the findings entry annotated 3/3 seeds.
- **48c99128** — a read-only validity check prompted by a reader objection to this page's 0.906 figure (internal run log #50). Bit-identity verified directly; the estimator's behaviour derived in closed form. No published number moved and no confirmed-findings entry changed; the methodological paragraph in §4 was added as a result.

Audit status: audit pending. This program requires every promoted finding to be independently audited by a party that did not run the science, re-deriving its numbers from the raw result files with freshly written code. That audit has not yet been carried out for either of the two findings reported

here. The 2026-08-12 validity check recorded above is narrower and was run inside the program; it is not that audit and does not stand in for it. Replication and audit are different guarantees, and this result currently has the first and not the second.

All claims judged strictly against the registered wording, misses included above; 3 seeds; replicated before publication. Internal designation: orthogonal reservoir.

AuxiLab · findings are pre-registered and replicated before publication

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